

Gal Arnon

Postdoctoral Researcher · Bocconi University

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Research Interests

I am broadly interested in cryptography and complexity theory. My research focuses on probabilistic proof systems (e.g., PCPs and IOPs) as well as their cryptographic counterparts, with an emphasis on succinct non-interactive arguments. I study both the theoretical foundations and practical aspects of these systems.

Appointments

2025–present	Postdoctoral Researcher, Bocconi University Milan, Italy. Hosted by Prof. Alon Rosen.
2025	Research Fellow, Simons Institute for the Theory of Computing Berkeley, USA.

Education

2020–2025	Ph.D. Computer Science, Weizmann Institute of Science Advisors: Prof. Moni Naor and Dr. Eylon Yogev. Thesis: <i>New Advancements in Interactive Oracle Proofs: Theory, Practice, and Limitations.</i>
2017–2020	M.Sc. Computer Science, Weizmann Institute of Science Advisor: Prof. Guy N. Rothblum. Thesis: <i>On Prover-Efficient Public-Coin Emulation of Interactive Proofs.</i>
2013–2017	B.Sc. Electrical Engineering & Computer Science, Tel Aviv University <i>Magna Cum Laude.</i>

Awards

2024	Best Paper Award, CRYPTO 2024 For “STIR: Reed–Solomon Proximity Testing with Fewer Queries”.
2024	Esther Hellinger Memorial Prize for Academic Excellence, Weizmann Institute of Science

Papers

Preprints

2. **Interactive Proofs for Batch Polynomial Evaluation**
Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev
Manuscript
1. **Pairing-Based SNARGs with Two Group Elements**
Gal Arnon, Jesko Dujmovic, and Eylon Yogev
Manuscript

Publications

11. **Designated-Verifier SNARGs with One Group Element**
Gal Arnon, Jesko Dujmovic, and Yuval Ishai
CRYPTO 2025
10. **Towards a White-Box Secure Fiat-Shamir Transformation**
Gal Arnon and Eylon Yogev
CRYPTO 2025, ZKProof 7.
9. **Instance Compression, Revisited**
Gal Arnon, Shany Ben-David, and Eylon Yogev
EUROCRYPT 2025
8. **WHIR: Reed–Solomon Proximity Testing with Super-Fast Verification**
Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev
EUROCRYPT 2025, ZKSummit 12, ZKProof 7.
7. **Hamming Weight Proofs of Proximity with One-Sided Error**
Gal Arnon, Shany Ben-David, and Eylon Yogev
TCC 2024
6. **STIR: Reed–Solomon Proximity Testing with Fewer Queries**
Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev
CRYPTO 2024, ZKSummit 11, ZKProof 6. **[Best Paper Award]**
5. **IOPs with Inverse Polynomial Soundness Error**
Gal Arnon, Alessandro Chiesa, and Eylon Yogev
FOCS 2023, ITC 2024 Highlights
4. **A Toolbox for Barriers on Interactive Oracle Proofs**
Gal Arnon, Amey Bhangale, Alessandro Chiesa, and Eylon Yogev
TCC 2022
3. **Hardness of Approximation for Stochastic Problems via Interactive Oracle Proofs**
Gal Arnon, Alessandro Chiesa, and Eylon Yogev
CCC 2022
2. **A PCP Theorem for Interactive Proofs and Applications**
Gal Arnon, Alessandro Chiesa, and Eylon Yogev
EUROCRYPT 2022

1. On Prover-Efficient Public-Coin Emulation of Interactive Proofs

Gal Arnon and Guy N. Rothblum

ITC 2021

Media Coverage

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| 2025 | Quanta Magazine
Computer Scientists Figure Out How to Prove Lies mentions our work “Towards a White-Box Secure Fiat-Shamir Transformation”. |
| 2024 | ZK Podcast, Episode 321
Interview on the STIR protocol. |

Selected Talks

- **Fiat-Shamir Attacks and Mitigations**
 - Proofs Workshop, “Cryptography, 10 Years Later” Semester, Simons Institute, Berkeley, USA. *July 2025.*
 - Proofs Reading Group, “Cryptography, 10 Years Later” Semester, Simons Institute, Berkeley, USA. *June 2025.*
 - NTT Cryptography Seminar (Virtual). *April 2025.*
- **Instance Compression, Revisited**
 - Foundations Reading Group, “Cryptography, 10 Years Later” Semester, Simons Institute, Berkeley, USA. *July 2025.*
- **STIR: Reed–Solomon Proximity Testing with Fewer Queries**
 - CRYPTO 2024, Santa Barbara, USA. *August 2024.*
 - Interuniversity TCS Student Seminar, Tel Aviv University, Israel. *May 2024.*
 - Theory Lunch, Weizmann Institute of Science, Israel. *May 2024.*
 - HUJI TCS Seminar, Jerusalem, Israel. *May 2024.*
 - StarkWare Industries, Netanya, Israel. *April 2024.*
 - ZKSummit 11, Athens, Greece. *April 2024.*
- **IOPs with Inverse Polynomial Soundness Error**
 - ITC 2024 Highlights Track, Stanford University, USA. *August 2024.*
 - Technion TCS Seminar, Haifa, Israel. *February 2024.*
 - ZK Study Club (Virtual). *October 2023.*
 - StarkWare Industries, Netanya, Israel. *September 2023.*
 - Interuniversity TCS Student Seminar, Tel Aviv University, Israel. *July 2023.*
 - IST Austria TCS Seminar, Vienna, Austria. *June 2023.*
- **How To Be Convinced While Barely Listening (Even to Yourself)**
 - Theory Lunch, Weizmann Institute of Science, Israel. *May 2025.*
 - EPFL CS Theory Reading Group, Lausanne, Switzerland. *May 2023.*
 - Efficient Probabilistic Proofs Workshop, Bertinoro, Italy. *July 2022.*
- **A Toolbox for Barriers on Interactive Oracle Proofs**
 - TCC 2022, Chicago, USA. *November 2022.*
- **Hardness of Approximation for Stochastic Problems via Interactive Oracle Proofs**
 - CCC 2022, Philadelphia, USA (Virtual). *July 2022.*

- **A PCP Theorem for Interactive Proofs and Applications**
 - EUROCRYPT 2022, Trondheim, Norway. *May–June 2022.*
 - Theory Lunch, Weizmann Institute of Science, Israel. *July 2021.*
- **On Prover-Efficient Public-Coin Emulation of Interactive Proofs**
 - ITC 2021 (Virtual). *July 2021.*
 - “Proofs, Consensus, and Decentralizing Society” Program Seminar, Simons Institute, Berkeley, USA. *October 2019.*

Service

Proximity Prize	Committee member and judge for the Proximity Prize initiative of the Ethereum Foundation — a bounty to prove or disprove Reed–Solomon proximity gap conjectures.
Workshop	Lattices Meet Hashes: Recent Advances in Post-Quantum Zero-Knowledge Proofs . Postdoctoral workshop, Bernoulli Center, EPFL, Lausanne. Co-organized with Ngoc Khanh Nguyen. May 2023.
Program Comm.	TCC 2026
Reviews	CCC (2024) · CRYPTO (2019, 2022, 2023, 2024, 2025, 2026) · EUROCRYPT (2023, 2026) · FOCS (2025) · ITC (2025) · ITCS (2022, 2026) · SICOMP (2026) · SODA (2024) · STOC (2025) · TCC (2021, 2023, 2025)

Teaching

July 2023	Teaching Assistant , <i>Foundations and Frontiers of Probabilistic Proofs</i> MSRI (SLMath) Summer Graduate School, ETH Zurich, Switzerland.
July–Aug 2021	Teaching Assistant , <i>Foundations and Frontiers of Probabilistic Proofs</i> MSRI (SLMath) Summer Graduate School, Virtual.
Sept 2018	Instructor , <i>Mini-Course on Zero-Knowledge Proofs</i> Amos de-Shalit Summer School, Weizmann Institute of Science.